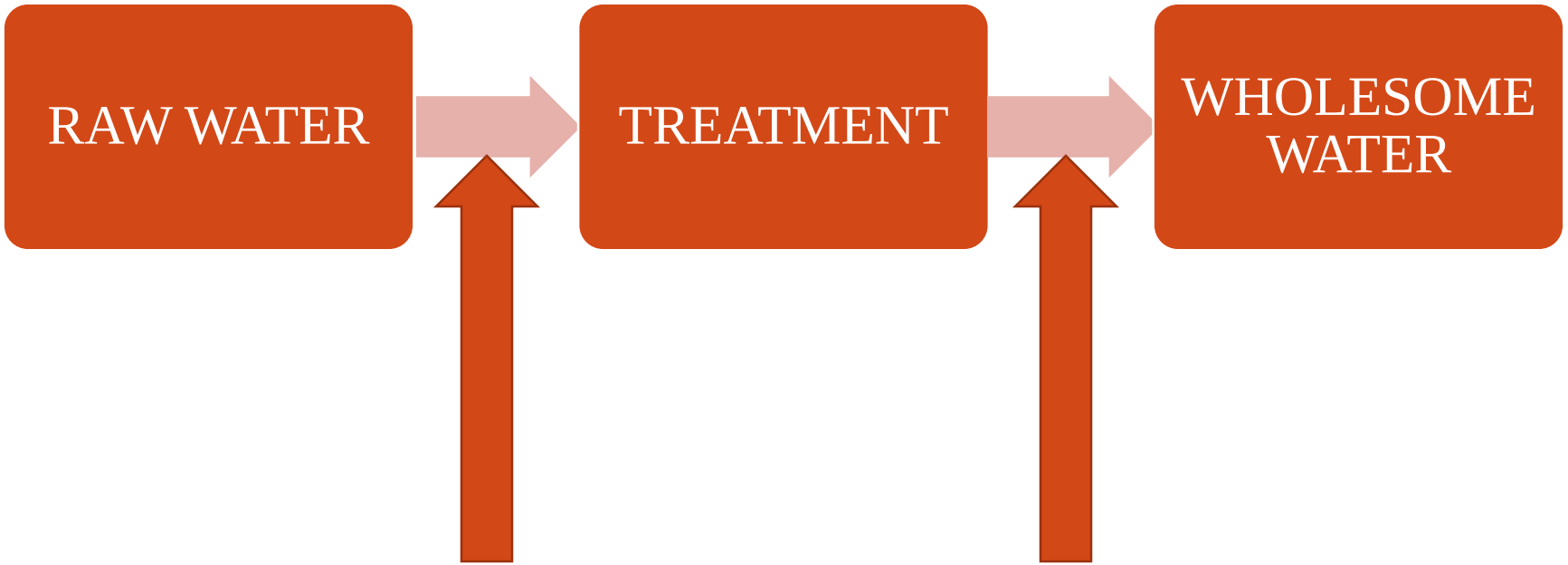
Water Quality



Water Quality

- Water is one of the most important material required to sustain life & is considered as the source of human illness.
- It is rarely available in nature because it has great tendency to dissolve various substances in it while flowing on the surface of earth.
- The topic of water quality focuses on the presence of foreign substances in it & their effect on human & other aquatic life.
- **Where as WHO listed over 100 chemicals in guide lines for drinking water.**
- In addition, the public need water that is soft, non-corrosive, free from pathogens & suspended solids.
- Due to inferior quality of water millions of children die every year. In the world under the age of 5years, due to water born diseases.
- Water quality for one purpose is not good / suitable for other purpose.

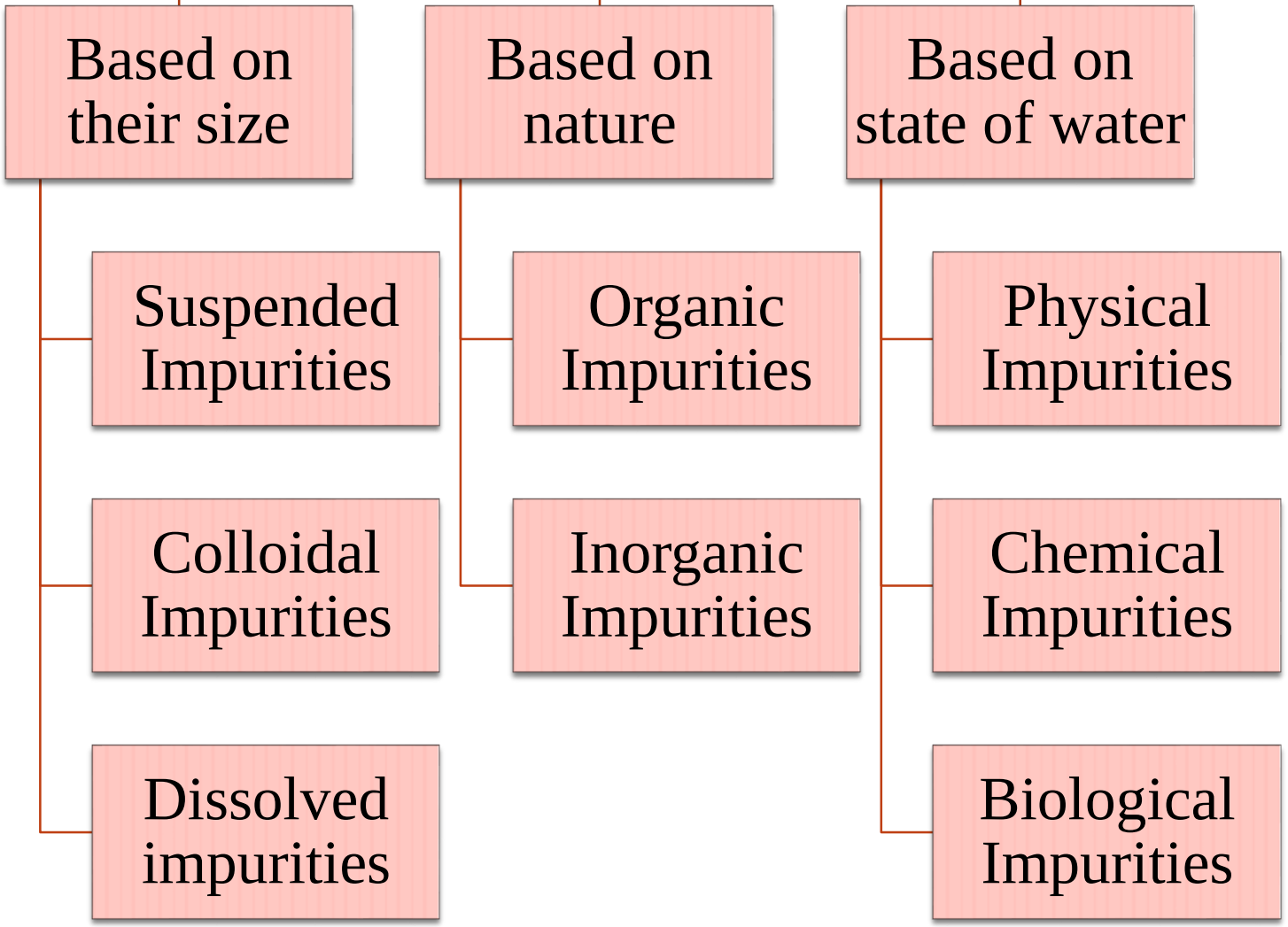
IMPORTANT REQUIRMENTS OF WATER FOR DOMESTIC USE:

- It should be colorless and sparkling water
- It should be of good taste , free from odour.
- It should be reasonably soft.
- It should be free from disease producing bacteria or pathogens.
- It should be free from objectionable dissolved gases.
- It should be free from harmful salts.
- It should be free from objectionable minerals such as iron , manganese ,lead, arsenic and other poisonous metals.
- It should be free from radio active substance such as radium,stronsium etc.
- It should not lead to scale formation and should be non-corrosive.

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THE ABOVE REQUIRMENTS ARE ONLY QUALITATIVE.

CLASSIFICATION OF IMPURITES



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CLASSIFICATION OF IMPURITES BASED ON THEIR SIZES

Suspended

- Particle size is in between $10^{-1}mm\ to\ 10^{-3}mm$
- Normally remain in suspension
- They are microscopic and make water turbid

Colloidal

- Particle size is in between $10^{-3}mm\ to\ 10^{-6}mm$
- This are electrically charged
- Due to this this are very small in size remains in constant motion

Dissolved

- Particle size is in between $10^{-6}mm\ to\ 10^{-8}mm$
- This are not visible to naked eyes.
- They causes bad taste, hardness and alkalinity

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Type	Constituents	Effect
1. Suspended Impurities	(a) Bacteria	Some cause diseases
	(b) Algae, Protozoa	Odour, colour, turb
	(c) Silt	Murkiness or turbi
2. Dissolved Impurities	(a) *Salts*	
	(i) Calcium and Magnesium — Bicarbonate	Alkalinity
	(i) Calcium and Magnesium — Carbonate	Alkalinity, hardness
	(i) Calcium and Magnesium — Sulphate	Hardness
	(i) Calcium and Magnesium — Chloride	Hardness, corrosion
	(ii) Sodium — Bicarbonate	Alkalinity, softening
	(ii) Sodium — Carbonate	Alkalinity, softening
	(ii) Sodium — Sulphate	Foaming in boilers
	(ii) Sodium — Fluoride	Dental flurosis or m led enamel
	(ii) Sodium — Chloride Taste	
	(b) *Metals and Compounds*	
	(i) Iron oxide	Taste red colour, c siveness, hardness
	(ii) Manganese	Black or brown colou
	(iii) Lead	Cumulative poisoning
	(iv) Arsenic	Toxicity, poisoning
	(v) Barium	Toxic effect on he nerves
	(vi) Cadmium	Toxic, illness
	(vii) Cyanide	Fatal
	(viii) Boron	Affect central nerv system
	(ix) Selenium	Highly toxic to anim fish
	(x) Silver	Discolouration of sk eyes
	(xi) Nitrates	Blue baby condition infant poisoning colo acidity
	(c) *Vegetable dyes*	
	(d) Gases — Oxygen	Corrosiveness to metal
	(d) Gases — Carbondioxide	Acidity, corrosiveness
	(d) Gases — Hydrogen sulphide	Odour, acidity, cor siveness

WATER ANALYSIS

- Various types of impurities present in water can be determined by water analysis.
- This analysis is done both for raw water as well as treated or purified water.
- The examination of raw water will be enable us to determine the outline or processes of water purification.
- Water analysis of purified water is done to know whether the degree of purification has reached the required standards or not.
- Examination of water is used to classify,presribe treatment , control treatment and purification processes and maintain public supplies of an appropriate standard of organic quality, clarity.

CLASSIFICATION OF IMPURITES BASED ON STATE OF MATTER

Physical Parameter

- Temperature
- Color
- Turbidity
- Odor and taste

Chemical Parameter

- Total solids
- Chlorides
- Hardness
- pH value
- Metal and other chemical Substances
- Alkalinity
- Nitrogen and its compounds
- Dissolved gases

Biological Parameter

Qualitative Test :

Plate count method and membrane filter method

Quantitative test:

E-coli test –

a)E-coli index

b)Most probable number

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Physical Parameters of Water Quality

1. Temperature:

CHEMICAL WATER QUALITY DEPENDS ON TEMP.

$$Growth\ of\ microorganism\ \alpha\ Temperature$$

$$Solubility(presence\ of\ gases\ in\ water)of\ gases\ \alpha\ \frac{1}{temperature}$$

$$Viscosity\ of\ water\ \alpha\ \frac{1}{temperature}$$

DRINKING WATER STANDARD w.r.t Temperature : 10º C to 20 º C

2. Color:

- color of water is due to presence of colloidal matter, decaying vegetation.
- Color in water is measured by a device known as “ TINTOMETER” on burgage scale (Platinum cobalt scale) in terms of true color units(TCU)
- scale of color from 0 to 70 units.

1 TCU = 1 mg of platinum as a chloroplatnium iron mixed in 1 liter. Of distilled water , the color produced is taken as 1 TCU (mg/ltr)

DRINKING WATER STANDARD w.r.t COLOR : 5 TCU to 20 TCU

3. Turbidity:

- It is measure of resistance offered by the particle present in water to the passage of light through water.
- **It is caused by suspended or colloidal particles.**
- Turbidity in water is measured on silica scale and expressed in terms of turbidi units.
- **Turbidity in water measured by device works on the light absorption.**

$$Turbidity \propto Surface\ area\ of\ particle \propto \frac{1}{size\ of\ particles}$$

1 Std turbidi unit = 1 mg of finely devided silica mixed in 1 ltr of distilled water, turbidity is taken as 1 STU (JTU).
THIS IS PROPOSED BY JACKSON

Method	Used for
Jackson Turbidity Rod	Field Measurement
Jackson Turbidity Meter	Measurement turbidity of water > 25 JTU
Baylis Meter	Low turbidity in water (up to 5 JTU)
Hilliage Meter	Measurement turbidity of water up to 50 JTU

- Device which works on principle of light scattering is known as NEPHLEOMETER.
- It measured only 90 degree scattering.
- 1 NTU = 1 JTU

DRINKING WATER STANDARD w.r.t TURBIDITY : 5 TO 10 JTU

4.Odour and test :

- **Odor is measured by device known as OSMOSCOPE.**
- **It is expressed in terms of Threshold Odor Number.**
- It is dilution ratio at which odor is just detectable.
- If odor is detected in water then we presumed that water also taste bad.

$$TON = \frac{A+B}{A}$$

where A = Volume of water sample tested (ml)

B= Volume of distilled water used in dilution (ml)

DRINKING WATER STANDARD w.r.t ODOUR:
TON < 3

Chemical Parameters of Water Quality

Many organic & inorganic chemicals affect water quality. In drinking water these chemicals affects public health, where as in surface water it affect to the aquatic life. Most important chemicals are,

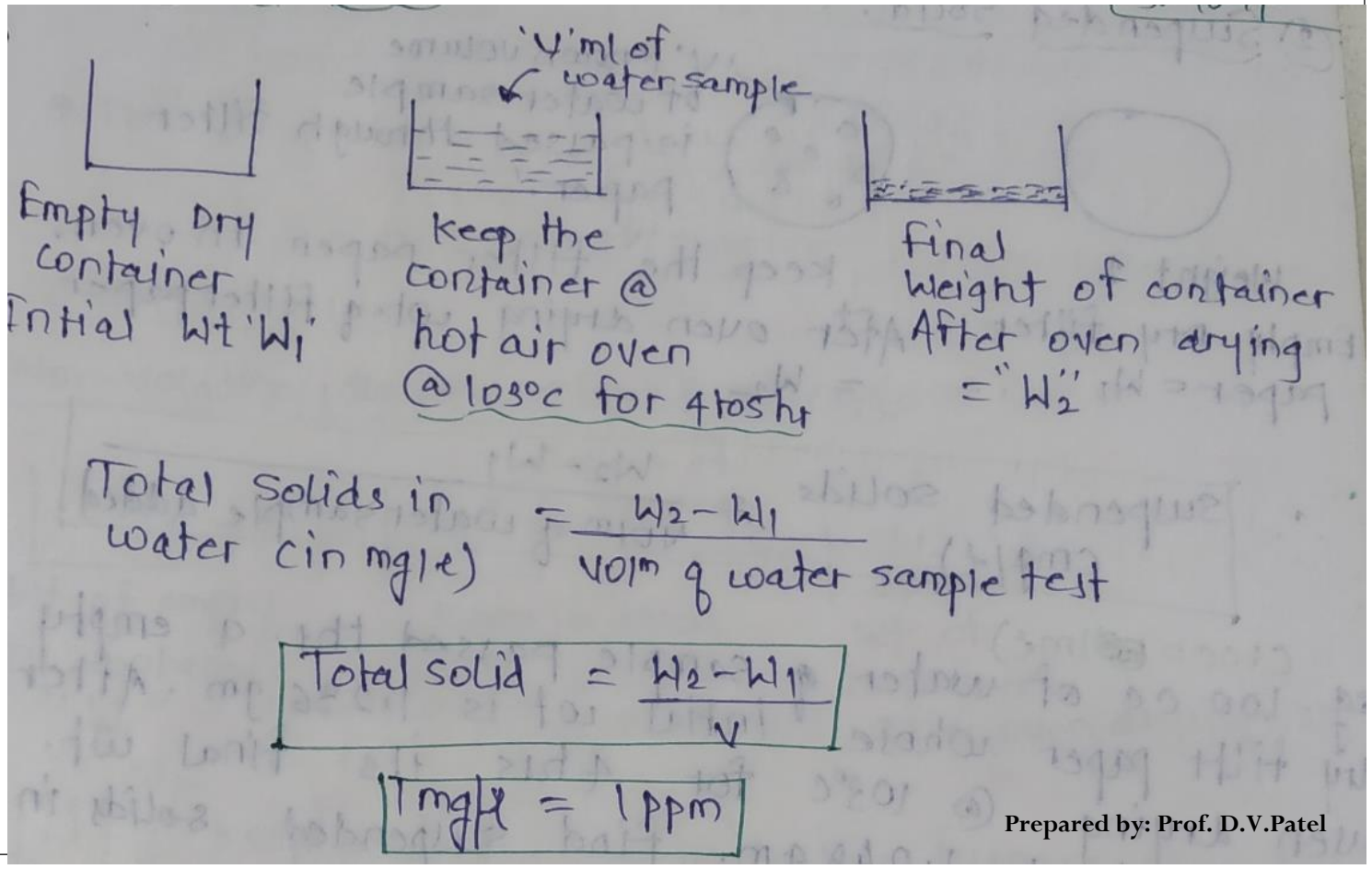
1. Acidity, Alkalinity & pH
2. Hardness
3. Total Dissolved Solids (TDS)
4. Toxic Chemicals
5. Dissolved Oxygen (DO)
6. Fluorides
7. Chlorides
8. Sulfates
9. Nitrates
10. Phosphates

1.Solids: Residue left on evaporation

Solids in water is measured by ‘Gravimetry’.

A. Total Solid
B. Suspended Solid
C. Total Dissolved Solid (TDS)

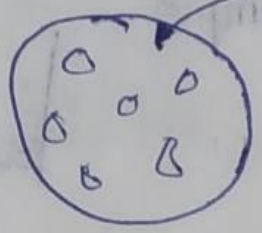
Total Solid



(2) Suspended solid:



Weight 1
Empty dry filter paper = W_1



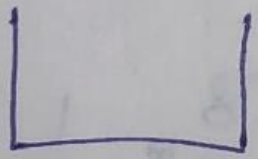
'V' known volume of water sample is passed through filter paper.

keep the filter paper in oven.
After oven drying wt. of filter paper $= W_2$

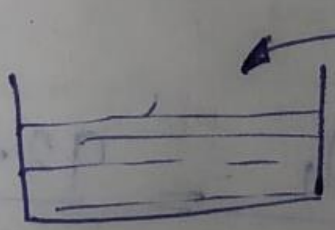
$$\therefore \text{Suspended solids (mg/l)} = \frac{W_2 - W_1}{\text{Vol}^m \text{ of water sample added}}$$

(3) Total Dissolved Solids:

TDS = Colloids + dissolved.



Initial wt. of empty container = W_1



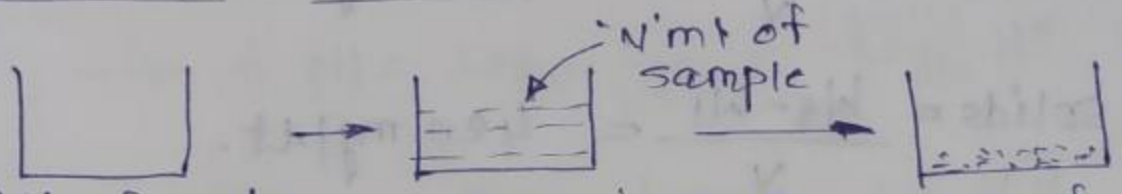
'V' ml of filtered water sample tested

After oven drying = W_2

$$TDS = \frac{W_2 - W_1}{\text{Volm of water 'V'}}$$

Volatile Solids:

Non-Volalite Solids: (fixed solids):



Wt. of empty dry container "W_1"

Keep in oven then Wt. = "W_2"

Wt. of container after oven drying = W_2



keep the container in muffle furnace @ 700°C - 800°C for 2 hr



Fixed Wt. of container "W_3"

$$\text{Volatile Solids} = \frac{W_2 - W_3}{V} \quad (\text{mg/l})$$

$$\text{Non-Volatile Solids} = \frac{W_3 - W_1}{V}$$

* pH : (potential hydrogen ion concentration)

$H_2O \longrightarrow H^+ + OH^-$ — Ionization

$$\frac{\text{Products}}{\text{Reactants}} = K$$

$$\therefore \frac{[H^+][OH^-]}{[H_2O]} = K$$

for pure water $K = 10^{-14}$ mol/lit, $H_2O = 1$

$$\therefore \frac{[H^+][OH^-]}{1} = 10^{-14}$$

$$\therefore [H^+][OH^-] = 1 \times 10^{-14} \text{ mol}^2/\text{lit}^2$$

Apply $\log_{10}$ on both side

$$\log_{10} H^+ + \log_{10} OH^- = -14$$

$$\log_{10} \frac{1}{H^+} + \log_{10} \frac{1}{OH^-} = 14$$

$$\therefore \boxed{pH + pOH = 14}$$

a) $pH = \log_{10} \frac{1}{H^+} = -\log_{10} H^+$

b) $pOH = \log_{10} \frac{1}{OH^-} = -\log_{10} OH^-$

eg. If pH of water is 6.5 then $[H^+]$ & $[OH^-]$ = ?

→ $pH + pOH = 14$

$pOH = 14 - 6.5$

$pOH = 7.5$

∴ ~~$= -\log_{10} H^+ = -\log_{10}$ [illegible]~~

$pOH = -\log [OH^-]$ $\quad pH = -\log [H^+]$

$7.5 = -\log_{10} [OH^-]$ $\quad 6.5 = -\log [H^+]$

∴ $\boxed{[OH^-] = 10^{-7.5} \text{ mol/Lt}}$ ∴ $\boxed{[H^+] = 10^{-6.5} \text{ mol/L}}$

OR

$pH = \log_{10} \frac{1}{H^+}$

$6.5 = \log_{10} \frac{1}{H^+}$

$\frac{1}{H^+} = 10^{6.5}$

∴ $\boxed{H^+ = 10^{-6.5} \text{ mol/Lt}}$ ∴ $OH^- = 10^{7.5} \text{ mol/Lt}$

(22/36) $pH = 4.1$, $[H^+] = ?$

$\therefore [H^+] = 10^{-4.1}$ mol/Lit

$\boxed{[H^+] = 7.94 \times 10^{-5} \text{ mol/Lit}}$

(37) $OH^- = 10^{-5.6}$ mmol/Lit ~~pOH = 5.6~~

$= 10^{-5.6} \times 10^{-3}$ mol/Lit

$\therefore OH^- = 10^{-8.6}$ mol/Lit

$\therefore pOH = 8.6$

$\therefore pH = 14 - 8.6 = 5.4$

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Application of pH:

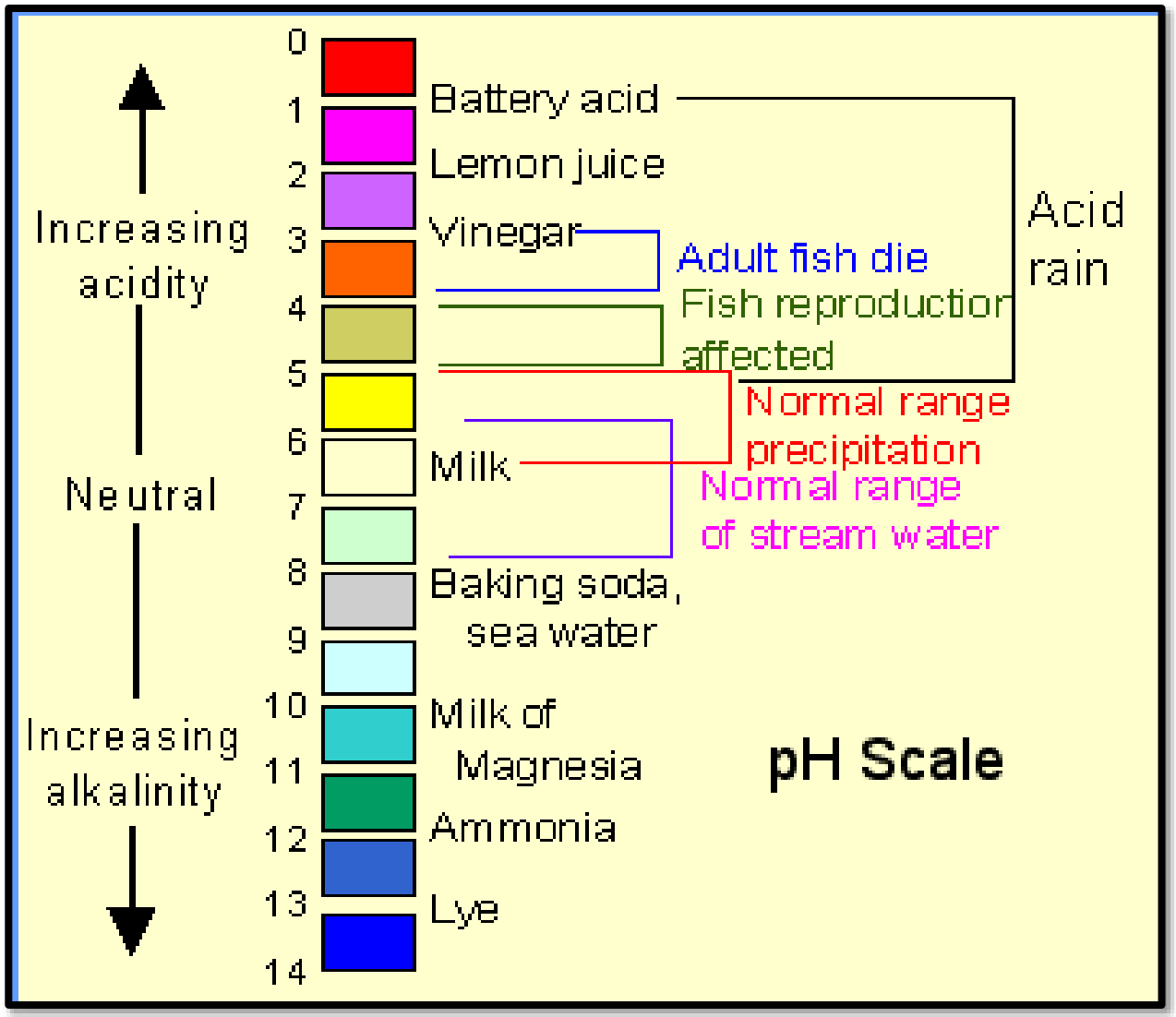
- Water treatment is pH is sensitive.
- **Dosage of chemical depends on Ph.**
- Extreme pH value damage property.
- Nature of water can be determined by pH.

Ph in water is measured by:

- pH meter
- Titrometer
- Potentiometer (pH meter)

DRINKING WATER STANDARD w.r.t pH:
6.5 to 8.5

pH







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Acidity: Measure of ability of water to neutralize base.

- Mineral Acidity = pH – 0 to 4.2
- Carbonic Acidity = pH – 4.2 to 8.2

Disadvantages of acidity:

1. Corrosion
2. Tuberculation – Interior of pipe becomes rough

- Water used in construction = acidity < 50 mg/l as CaCO3

Alkalinity: Measure of ability of water to neutralize Acid.

Alkalinity Caused due to :

1. OH^- = pH > 10.2
2. CO_3^- = pH = 8.2 to 10.2
3. HCO_3^- = pH = 4.2 to 8.2

Disadvantages of Alkalinity :

1. Incrustation is deposition of flowing fluid on the surface of pipe decreasing the cross section area of pipe and for same velocity of flow may lead to reduced discharge
2. Bitter Taste

- Alkanity in water expressed in mg/l as $CaCO_3$

"x" in terms of "y" mg/l = $\frac{x}{(mg/L)} * \frac{\text{Eq. wt. of } y}{\text{Eq. wt. of } x}$

Total alkanity (TA) = $\frac{OH^-}{(mg/l)} * \frac{\text{Eq. wt. of } CaCO_3}{\text{eq. wt. of } OH^-}$

(M) T.A. in mg/l as $CaCO_3$ = $\frac{OH^-}{(mg/L)} * \frac{50}{17}$

T.A. in mg/l as $CaCO_3$ = $\frac{CO_3^{--}}{(mg/l)} * \frac{\text{Eq. wt. of } CaCO_3}{\text{eq. wt. of } CO_3^{--}} + HCO_3^- * \frac{\text{Eq. wt. of } CaCO_3}{\text{Eq. wt. of } HCO_3}$

(M) T.A. in mg/l. as $CaCO_3$ = $\frac{CO_3^{--}}{(mg/l)} * \frac{50}{30} + \frac{HCO_3^-}{(mg/l)} * \frac{50}{61}$

Eq. wt. of $OH^- = \frac{16+1}{1} = 17$

Eq. wt. of $CO_3^{--} = \frac{12+3\times16}{2} = 30$

Eq. wt. of $HCO_3^- = \frac{1+12+3*16}{1} = 61$

Alkanity in water is measured by titrometry.

Titrent: H_2SO_4 diluted

Indicator: 1) phenolphthalin (pink-colourless) - palkanity

2) methyl orange (yellow-pink) m alkanity

(or) T alkanity

P & T relation	OH^-	CO_3^-	HCO_3^-
$P = 0$	0	0	T
$P < \frac{1}{2}T$	0	2P	T − 2P
$P = \frac{1}{2}T$	0	2P	0
$P > \frac{1}{2}T$	2P − T	2T − 2P	0
$P = T$	T	0	0

addition should be T

Hardness is a property of water, which represent the total concentration of Ca and Mg ions expressed as $CaCO_3$.

There are 2 forms of hardness. Carbonate hardness and Non-carbonate hardness.

- Carbonate hardness(Temporary) in natural waters is derived by the action of CO_2 (Which is a byproduct of microbial action on organic matter in top and subsoil) on limestone deposits. So soluble bicarbonates and carbonates of Ca and Mg contribute to carbonate hardness.
- Ones carbonates of limestone dissolved in waters acidified by CO_2, excess of Ca^{2+} and Mg^{2+} ions combined with available chlorides and sulphate to form non-carbonate hardness.(Permanat)
- Some hardness is also caused by cations Sr^{2+},,Fe^{2+}, and Mn^{2+}, excess of Ca^{2+} and Mg^{2+} ions combined with available chlorides and sulphates to form non-carbonates ions. SiO_2, but this is negligible and is relatively of less significance in water supply practice.

Types of hardness:

Range of hardness, mg/l as $CaCO_3$	Classification of hardness
0-50	Soft
50 - (75 - 100) – 150	Moderately soft
150 – 300	Hard
>300	Very Hard

Recommended standard for Hardness in drinking water, mg/l as CaCO3

Authority	HDL	MPL
BIS	300	600
GOI	200	600
WHO	100	500

Hardness in water is measured by Versanate Method (EDTA Method)

Disadvantages of Hardness :

1. Scaling
2. Corrosion
3. Bitter Taste
4. Increased soap consumption

Chlorides are present in all natural waters. Generally, chloride concentration is high if non-carbonate hardness is high.

- Fresh water sources can have chloride concentration varying from less than 100mg/l to more than 200mg/l, again like hardness; underground sources generally have higher chloride concentration than surface sources.

- Sea is the ultimate depositories of residues contain around 18000 to 20000 mg/l. of chlorides & are unpalatable & unfit for drinking & cause retching and purging, if consumed.

- Fresh water can derive high concentration of chlorides from: -

  a) Sewage effluents.
  b) Industrial effluents and
  c) Seawater infiltration.

Recommended standard for Chlorides in drinking water, mg/l as

Authority	HDL, mg/l	MPL, mg/l
BIS	250	1000
GOI	200	1000
WHO	200	600

Chlorides in water is measured by Mohr's Method

Chlorides in combination with sodium causes kidney and cardio problems

Importance of determination of chlorides: -

1)A chloride concentration of about 250mg/l is considered a limiting factor in selecting source of water for domestic supply.

2)When saline water is to be made potable, chloride concentration determines the method of desalination to be adopted.

3)A sudden increase in normal chlorides concentration in a domestic water supply may indicate possible sewage concentration.

4)Infiltration of seawater into neighboring fresh water wells can be controlled by frequent chloride determinations of well samples. yield of the wells can be restricted by limiting the chloride concentration to an acceptable level.

- **Sulphates** are very widely distributed in the earth's crust.
- Normal Sulphate characterized most of the metal ions and is mainly water-soluble and often crystallizes as hydrates. Excavation drainage wastes contribute high sulphates because of pyrite oxidation. Natural water contains varying amount of sulphates in them.
- Industrial wastewaters from pesticide manufacturing and formulation industries, paper and pulp industries etc. also contribute sulphates to receiving water bodies.
- Excessive dosing with alum (>450mg/l) during water treatment can introduce more than desirable concentration of sulphates in drinking water.

Recommended standard for Sulphate in drinking water, mg/l as

Authority	HDL	MPL
BIS	200	400
GOI	200	400
WHO	200	400

Sulphate in water is measured by Gravimetric method

Sulphate in High concentration causes Laxatic problems (Looses Motion-Diahrea)

Importance of Sulphates in Drinking Water:

1.Sulphates in water are insoluble at high temperatures and form scales in boilers and heat exchangers and result in wastage of fuel.

2.Sulphates In association with Ca and Mg, excessive sulphates produce high hardness and cause scale in cooking vessels and wastage of soap during washing.

3.Sulphates in the sewerage system are reduced to H2S under anaerobic condition and cause odour and corrosion problems.

4.Sulphates in anaerobic digester are reduced to sulphide and above pH=8, (S-) soluble sulphide concentration increases and in excess of 200mg/l gas production is inhabited. Below pH 7, H2S concentration in biogas increases and becomes unsuitable for use in gas engines.

Dissolved Oxygen (D.O.) levels in natural and wastewaters are dependent on the physical, chemical and biochemical activities prevailing in the water body.

- The analysis of D.O. is a key test in water pollution control activities and waste treatment process control.
- The presence of oxygen is essential for the survival of aquatic life in water. This oxygen is especially required by aerobic bacteria and other micro-organisms for degradation and stabilization of organic constituents in waste water.
- A rapid fall in DO level in river water is one of the first indications of organic pollution.
- Thus it is one of the important parameters for accessing the quality of water, water bodies and plays a key role in water pollution control activities.
- The major inputs of DO to natural water are from atmosphere and photosynthetic reaction. The solubility to oxygen in water depends on pressure, temperature, altitude and chloride concentration etc.
- Low DO in water can kill fish and many other organisms in water. For example, fish requires 2 mg/lt to 5 mg/lt of DO in water.

Dissolved oxygen in water is measured by Winkler's Method

DRINKING WATER STANDARD w.r.t DO: 3 to 4 mg/l

Environmental Significance of DO:

- The organic matter present in the waste water poses a Biochemical Oxygen Demand. This demand is met with the Dissolved Oxygen present in the fresh body of water
- Aerobic bacteria thrive when free oxygen is available in plenty. Aerobic conditions prevail when sufficient DO is available in water. End products of aerobic metabolism are stable and not foul smelling.
- Higher temperature, biological impurities, ammonia, nitrites, ferrous iron, chemicals such as hydrogen sulphide and organic matter reduces DO values.
- Algae group in water may release oxygen during its photosynthesis and DO may even shoot up to 300 mg/lt.
- Drinking water should be rich in DO for good taste. However higher value of DO in water may cause corrosion of iron and steel.
- DO test is necessary for all aerobic biological waste treatment process to control rate of aeration.

- DO test is used to evaluate the pollution strength of industrial and domestic waste.

Fluorides :-

(1) fluorides < 1mg/l → Dental cavities (or) Dental caries

fluorides + Enamel → Strong teeth

(2) fluorides > 1.5 mg/l Mottling of teeth (Teeth staining) yellow staining

9-11-14

(3) Fluorides excessively high:

Fluorors: Dental fluorors
Skeletal / Bone fluorides.

- fluorides in Drinking water estimated by colourometry by 'Zirconium'.

* Drinking water std w.r.t. fluorides: 1 to 1.5 mg/l

Nitrogen Compounds

(Indicates contamination of water with Sewage)

1. Ammonia nitrogen ⟶ Recent pollution
2. Organic Nitrogen (or) aluminoid nitrogen ⟶ old polln
3. Nitrites ⟶ Intermediate stage of decomposition
4. Nitrates ⟶ oxidized organic matter.

Kjeldahl nitrogen :

Nitrogen compounds are estimated by colorometry.

Drinking water std:

A.N. < 0.1 mg/lt

ON < 0.15 mg/lt

Nitrites = NIL (zero)

Nitrates < 45 mg/lt

(n) ⟶ Nitrates > 45 mg/lt ⟶ Methemoglobinaemia (Blue baby disease)

Prepared by: Prof. D.V.Patel

* Iron & magnese

- Iron cause radish colour to water
- Magenese — Brown to purple colour to water.
- Fe & Mn estimated by colourometry.

Drinking water std: Iron < 0.3 mg/l
magnese < 0.05 mg/l

[TM]

* Toxic Chemical Substances:
(Heavy metals)

Causes: 1) Destroy blood cells
2) damage blood vessels/ Central nervous System.
3) cardio problem
4) Commulative poisons.

Toxic substances:

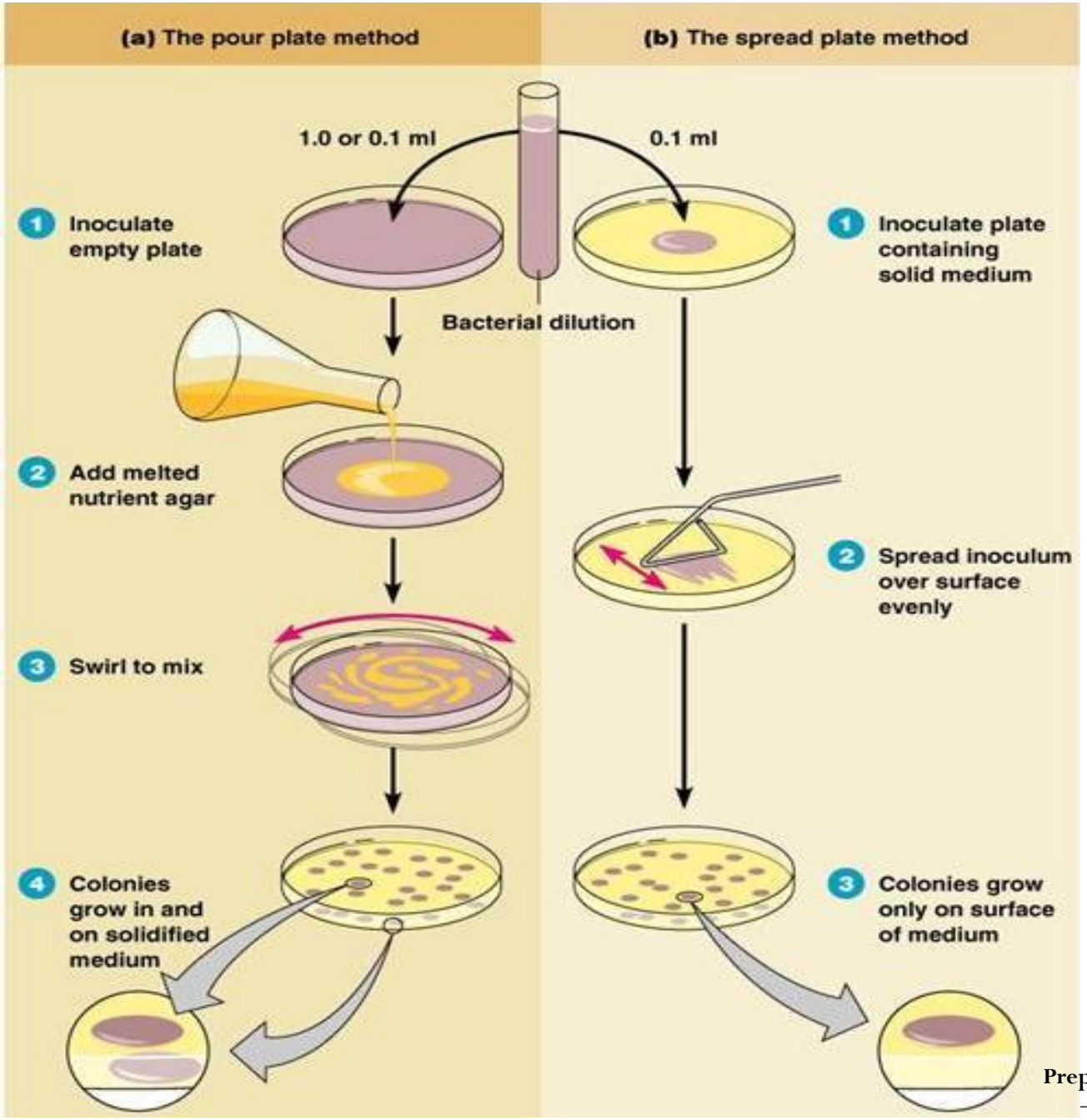
Lead	< 0.1 mg/l
Arsenic	< 0.05 mg/l
Barium	< 0.01 mg/l
Boron	< 1 mg/l
Cadmium	< 0.01 mg/l
Chromium	< 0.05 mg/l
Mercury	< 0.001 mg/l
phenol	< 0.001 mg/l
Cyanide	< 0.05 mg/l

Biological Parameters of Water Quality

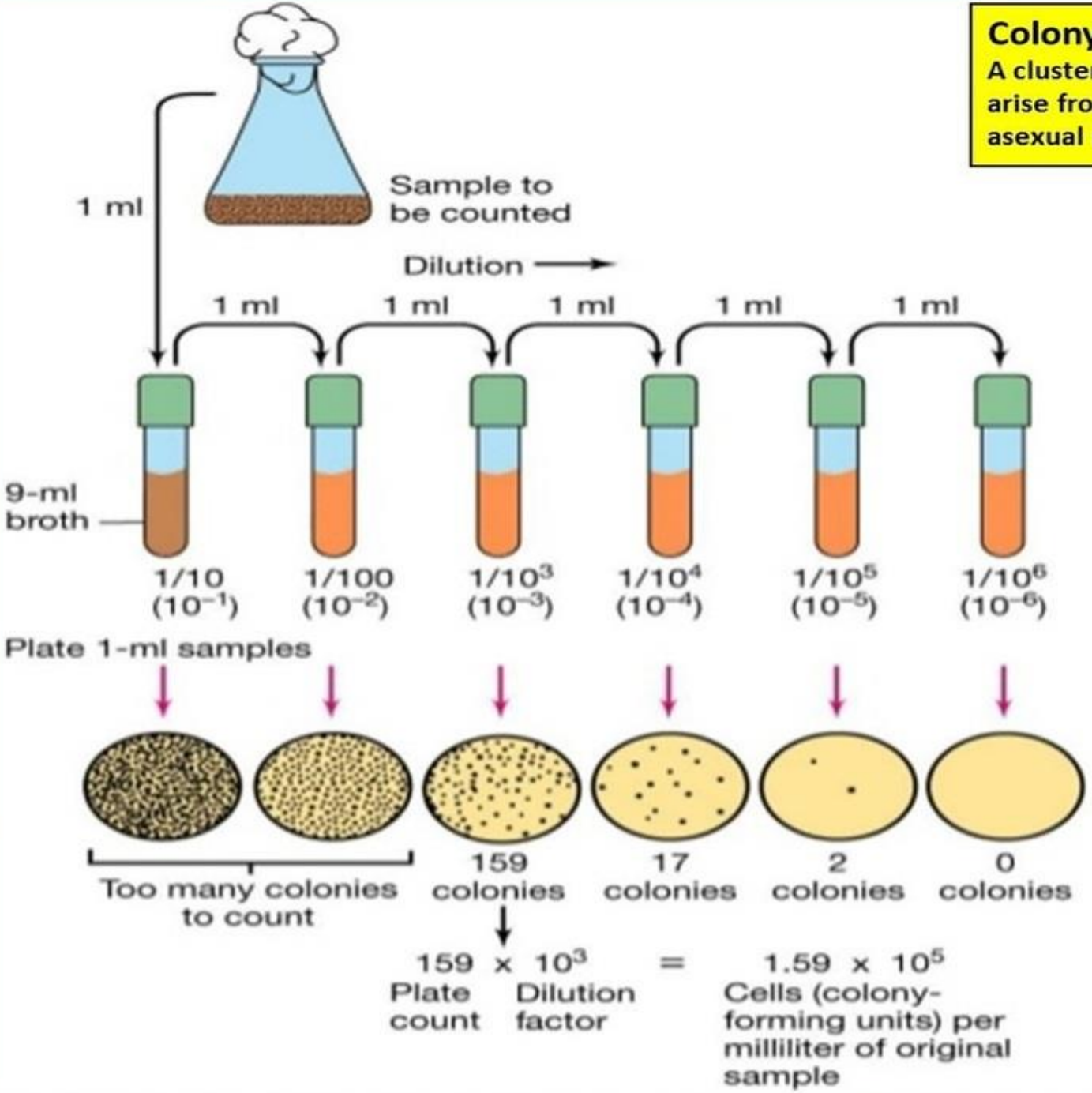
The presence & absence of living organisms in water can be one of the most useful indicator of the quality of water. The verity of species (Fishes) present in the lake be sign of unpolluted lake.

1. Micro-organisms
2. Bacteria
3. Algae
4. Protozoa
5. Viruses

- **Qualitative Test : Plate count method**



Colony of (bacteria):
A cluster of cells (or clones) which arise from a single bacterium by asexual reproduction



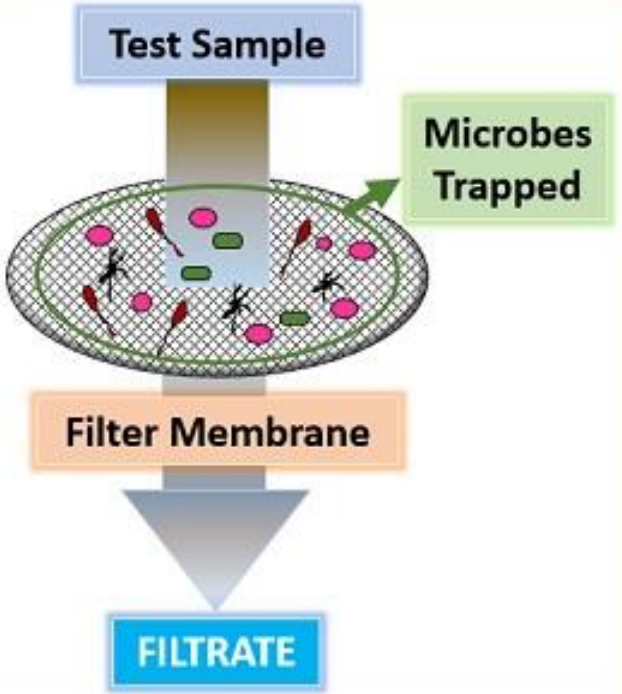
When a sample is plated, each colony that grows represents a single cell (or spore) in the original sample

Colony count multiplied by the dilution factor

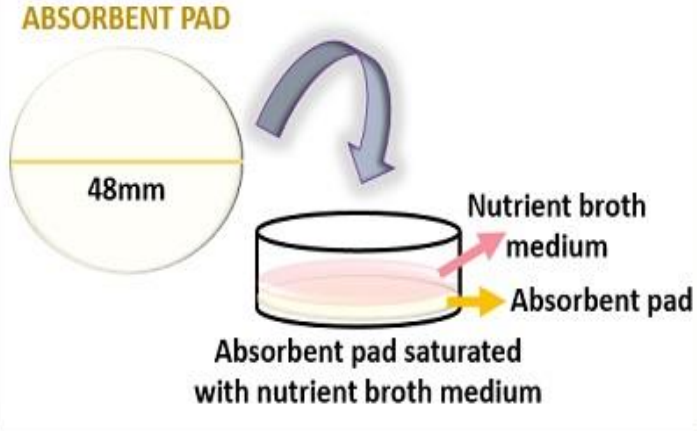
Prepared by: Prof. D.V.Patel

Membrane filter

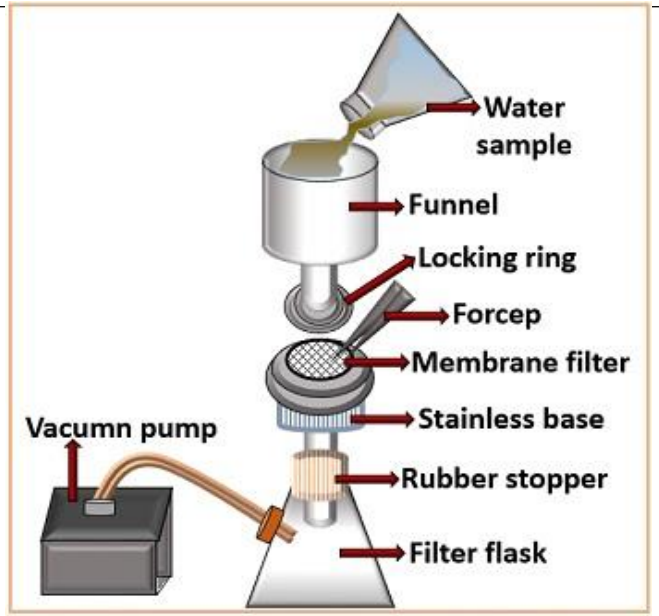
- It is a very effective method for the isolation and enumeration of microorganisms in the test water sample. By knowing the quantity of microbial mass, the quality of water can also determine.
- Therefore, the membrane filtration method can check both the quality of water and the quantity of microorganisms present in the water.
- It involves less preparation than many traditional **methods** and is one of a few **methods** that will allow the isolation and enumeration of microorganisms
- Membrane filtration method determines the microbial population by counting the number of discrete colonies, which is an advantage over the MPN method.
- In contrast to membrane filtration method, the MPN method determines the microbial density based on the turbidity of the culture broth.



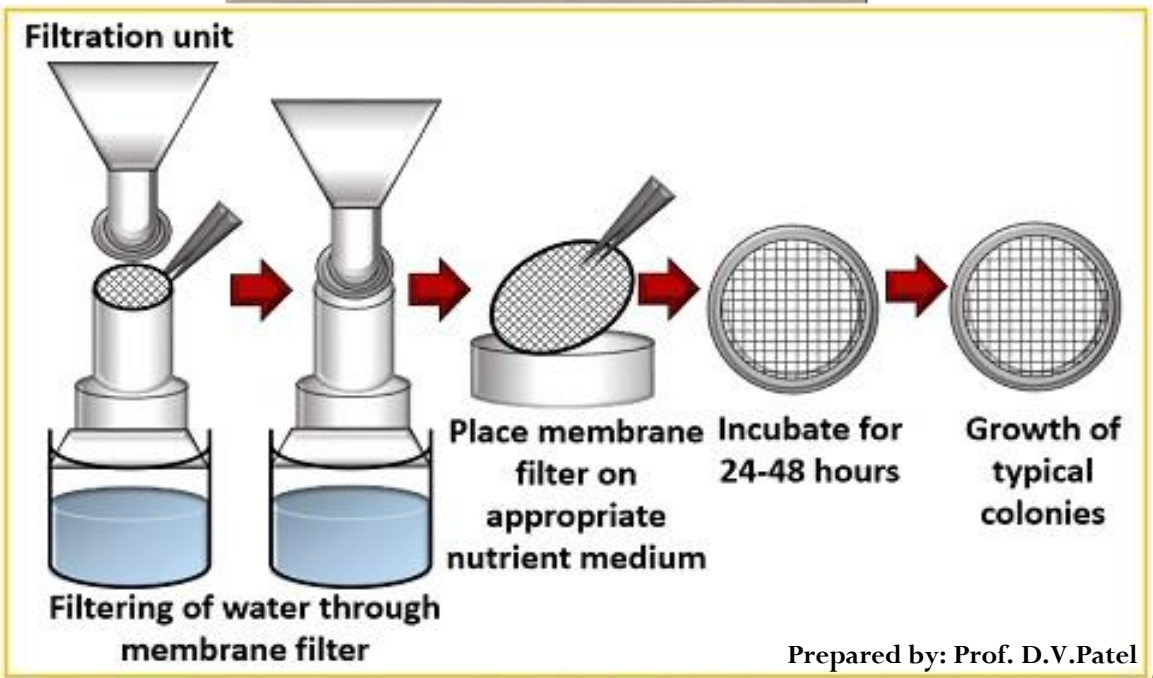
MEMBRANE FILTER



ABSORBENT PAD



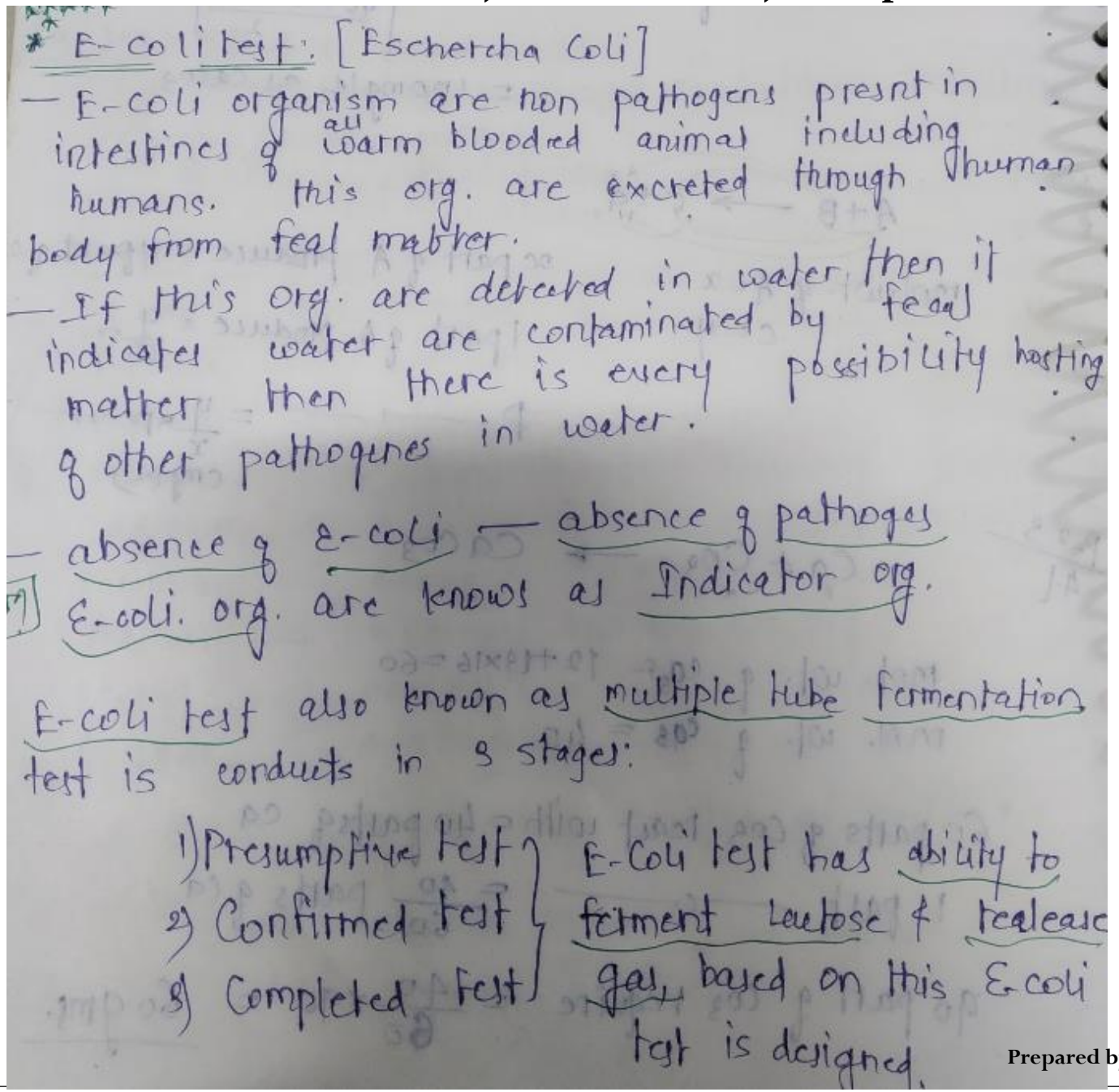
MEMBRANE FILTRATION UNIT



Prepared by: Prof. D.V.Patel

METHOD OF MEMBRANE FILTRATION

- **Quantitative test: E-coli test – a)E-coli index b)Most probable number**



* E-coli test: [Escherchia Coli]

— E-coli organism are non pathogens present in intestines of all warm blooded animal including humans. this org. are excreted through human body from feal matter.

— If this org. are detected in water then it indicates water are contaminated by feal matter then there is every possibility hosting of other pathogenes in water.

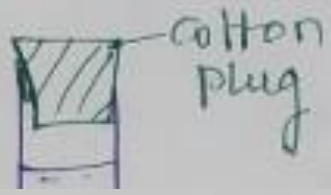
— absence of E-coli — absence of pathoges

E-coli. org. are knows as Indicator org.

E-coli test also known as multiple tube fermentation test is conducts in 3 stages:

1) Presumptive test
2) Confirmed test
3) Completed test

} E-Coli test has ability to ferment Lactose & realease gas, based on this E-coli test is designed.



9.11.19

E-coli Index:

Reciprocal of the smallest quantity of sample that would give +ve test results.

eg) 100ml, 10ml, 1ml, 0.1ml, 0.01ml, 0.001ml
+ve +ve +ve +ve -ve -ve

$$\therefore \text{E-coli Index} = \frac{1}{0.1} = 10 \text{ No/100ml}$$

2) 100ml, 10ml, 1ml, 0.1ml, 0.01ml, 0.001ml
+ + + − − −

$$\text{E-coli Index} = \frac{1}{1} = 1 \text{ no/100ml}$$

2. MPN : The laws of statistical probability are applied to the E-coli test results and Expressed in terms of MPN.MPN represent the bacterial density which is most likely to be present in the given water sample for the given test

Sample	No. tube tested	+ve	-ve
10	5	3	2
1	5	2	3
0.1	5	0	5

from MPN table for dilutn

+ve test result : (10-1-0.1)

MPN

5-5-5

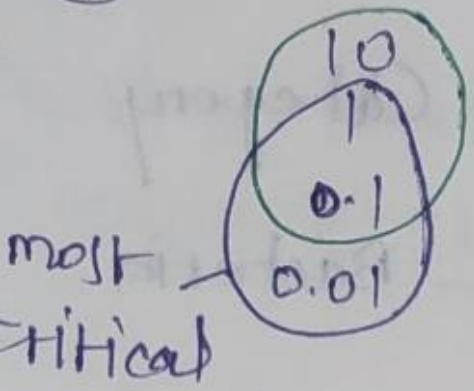
5-5-4

⋮

3-2-0 ——— 10

∴ MPN = 10 no/100ml

(4)



+ve	−ve
4	1
3	2
2	3
1	4

MPN table for 10-1-0.1

+ve

5-5-5

4-3-2 — 10

3-2-1 — 5 ✓

∴ MPN 1-0.1-0.01 = 5 * 10 = 50 mg/l.

* Drinking water std.:

E-coli index = 0

MPN ≯ 1 ND/100ml

≤

Water Borne Diseases :

	Name of the Disease	Organism	Category
1.	Cholera	Vibro cholerea (Vibro comma)	Bacteria
2.	Typhoid	Salmonela typhi	Bacteria
3.	Paratyphoid	Salmonela paratyphi	Bacteria
4.	Dysentery	Shigella dysenteria	Bacteria
5.	Amebiasis	Entamoeba & Hystolica	Protozoa
6.	Giardiasis	Giardia Lamblia	Protozoa
7.	Polio	Poliomyelitis	Virus
8.	Jaundice	Infectious Hepatitis	Virus